

Use Sentiment Analysis on Public Momentum to Predict Stock Trends

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Abstract

This capstone implements and evaluates a compact, reproducible pipeline that uses daily aggregated public sentiment (extracted with FinBERT) to predict next-day stock movement for Airbnb (ABNB). Addressing the resource gap between institutional and retail investors, the system scrapes daily ABNB-related text, computes day-level sentiment features (title/content probabilities), constructs 7-day rolling momentum features, and trains a logistic-regression classifier evaluated with temporally aware cross-validation. Results show a modest but detectable predictive signal (area under the ROC ≈ 0.55 in final tuning) with a logistic equation that reveals a significant reliance on headline sentiment over article content. Specifically, negative headlines were found to be the strongest predictor of downward price movement, validation of the "headline risk" hypothesis. The report documents design, methodology, implementation details (FastAPI pipeline), results, limitations, and recommendations for future work.

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Introduction

Background and Context

Retail investors rely heavily on publicly available text—news, blogs, and forums—to make trading decisions. Unlike institutional players who access proprietary data streams, retail traders operate in an environment defined by public sentiment. Prior market events, such as the GameStop short squeeze, illustrate how collective online sentiment can move markets, often decoupling price from fundamental value (Wikipedia Contributors, 2025). This project examines whether a compact, daily sentiment signal derived from readily available text can forecast short-term price direction for a single equity (Airbnb, ABNB).

Problem Statement

Many academic and industrial systems use large datasets and complex deep-learning models that are inaccessible to retail investors due to computational costs and data opacity. The practical gap addressed here is developing a low-overhead, reproducible pipeline that quantifies daily public sentiment and evaluates whether a seven-day sentiment momentum helps predict next-day rolling average price direction.

Objectives

1. Collect daily ABNB-related text and compute day-level sentiment via FinBERT.
2. Build 7-day rolling sentiment features and a logistic-regression model to predict next-day up/down movement.
3. Evaluate model via temporally oriented validation and report metrics, visualizations, and

interpretability analyses.

Scope

The project focuses on one stock (ABNB), daily data aggregation, FinBERT scoring, simple interpretable modeling (logistic regression), and reproducible pipeline documentation. It excludes intraday/high-frequency modeling, complex portfolio strategies, and "black box" deep-learning forecasting models.

Literature Review

The application of Natural Language Processing (NLP) to financial forecasting has evolved from simple lexicon-based approaches to sophisticated transformer-based architectures. This section synthesizes key theoretical frameworks and empirical studies that underpin this project's design.

The Evolution of Sentiment Analysis in Finance

Early attempts at financial sentiment analysis relied on "bag-of-words" approaches and general-purpose sentiment lexicons. However, Bharathi and Geetha (2017) demonstrated that while these methods could improve prediction over naive baselines, they struggled with the nuance of financial vernacular. Words like "liability" or "risk" may be negative in general conversation but neutral in a financial report. This limitation necessitated the move toward domain-specific models.

Domain-Adapted Transformers (FinBERT)

The introduction of BERT (Bidirectional Encoder Representations from Transformers) marked a paradigm shift. Genc (2020) and the ProsusAI (2020) repository highlight that fine-tuning BERT on financial corpora (FinBERT) significantly improves classification accuracy by capturing context specific to markets. Jiang and Zeng (2023) provided a systematic empirical study confirming that FinBERT-derived features consistently outperform classic LSTM and lexicon baselines in predicting stock movements. Their work serves as the central theoretical justification for this project's choice of FinBERT over simpler tools like VADER or TextBlob.

Furthermore, Shobayo et al. (2024) and Kirtac and Germano (2024) explore the frontier of using Large Language Models (LLMs) like GPT-4 for sentiment. While these models offer higher nuance, Kirtac and Germano warn of "model instability" and overfitting risks in retail trading scenarios. This validates the project's decision to use the stable, specialized FinBERT model rather than a generative LLM, balancing accuracy with computational reproducibility.

Temporal Dynamics and Hybrid Models

Recent literature emphasizes that sentiment is not static; it decays. Boyd (2019) reviews the lifecycle of news stories, suggesting that the "resonance" of a news item has a rapid initial spike followed by a tail. This supports the methodology's use of a 7-day rolling window, attempting to capture the lingering momentum of news rather than just instantaneous reactions.

While Gu et al. (2024) and Kim et al. (2023) demonstrate that feeding FinBERT vectors into LSTM (Long Short-Term Memory) networks yields superior predictive power, these architectures reduce interpretability. In contrast, this project aligns with the "Interpretable AI" framework discussed by Lundberg and Lee (2017). By feeding complex FinBERT features into a simple Logistic Regression, we maintain the ability to inspect coefficients directly—a critical requirement for retail investors who need to trust the signal.

Methodology

Approach

This project employs a quantitative predictive design: daily scraping → FinBERT sentiment scoring → 7-day rolling feature construction → logistic regression. The choice of a quantitative approach is justified by Streefkerk (2019), who notes that quantifying qualitative data (text) into a numeric series is essential for testing causal relationships in time-series data.

Data Sources and Collection Strategy

Following the principles of "Triangulation" (Bhandari, 2022), data were collected from multiple streams to reduce single-source bias:

- **Text:** Google News RSS, Yahoo Finance articles, and GDELT API.
- **Price:** Yahoo Finance via *yfinance* for ABNB OHLC data.

The scraping architecture was designed following Ališauskas (2025), treating the scraper as a stable "Data API." This involved implementing rate limiting, error handling, and standardized JSON output formats to ensure the pipeline could run robustly over the 8-week execution period.

Sentiment Extraction & Features

FinBERT was used to infer sentiment probabilities (Positive, Neutral, Negative) for both the Title and the Content of each article.

- **Aggregation:** Daily sentiment was computed by averaging the probabilities of all articles published between market close (previous day) and market open (current

day).

- **Imputation:** As noted by Mittal and Goel (n.d.), "thin-data days" (days with no news) create noise. To mitigate this, forward-fill imputation was used, assuming sentiment persists until new information arrives.

Modeling and Validation

- **Model:** Logistic Regression with L2 regularization.
- **Validation:** Walk-forward cross-validation was used to respect temporal order. Standard K-Fold CV was rejected as it would allow "future" data to leak into "past" training sets.

Project Plan and Execution

Development Environment & Architecture

The system was implemented as a FastAPI server (FastAPI + Uvicorn), Python 3.11. The modular architecture allowed for the separation of concerns:

- **/news**: Handles scraping and provenance logging.
- **/sentiment**: Wraps the FinBERT inference engine.
- **/predict**: Exposes the logistic regression model.

This design adheres to the operational best practices for data APIs outlined by Ališauskas (2025), ensuring that the system is not just a "script" but a deployable prototype.

Execution Summary & Schedule

Initial planning and system design were completed on schedule. However, FinBERT integration posed challenges. The original PyTorch weights caused compatibility issues on the development hardware (macOS CPU-only). Mitigation involved converting weights to a TensorFlow format and implementing batch processing to manage inference latency.

Challenges & Mitigations

The project encountered five distinct challenges during execution. Each was logged and mitigated to ensure the pipeline's reproducibility and validity.

Sparse or Uneven News Volume

- **Description**: Some trading days had very few or no ABNB-related articles. Because the model relies on day-level aggregated sentiment, these "thin-data days" reduce the

informativeness of daily features and risk introducing instability into the time series.

- **Impact:** This threatens feature consistency, making the sentiment series partially artificial. It increases the risk of feature–label misalignment and reduces the external validity of the model, as real-world trading algorithms depend on consistent signals.
- **Mitigation:** Forward-fill imputation was used to avoid missing values and preserve model compatibility. Daily source pools were broadened modestly (including blogs and GDELT) to increase coverage.

Forward-Fill Imputation Risk

- **Description:** Forward-filling sentiment replicates the previous day’s values when data is missing. While functionally necessary, it artificially smooths the sentiment curve.
- **Impact:** This reduces day-to-day volatility, potentially masking sentiment shocks. It acts as an undocumented "hidden model" inside the features, introducing structural bias.
- **Mitigation:** Forward-fill was chosen as the safest method for this dataset size (removing days would break the time series). The effect was monitored during exploratory analysis, and sentiment was left in probability form rather than discrete categories to minimize compounding bias.

FinBERT Environment and Compatibility Issues

- **Description:** FinBERT’s original PyTorch implementation produced environment

conflicts on the development hardware (macOS, CPU-only). Repeated dependency resolution attempts caused delays in Weeks 3–4.

- **Impact:** This introduced schedule risk and slowed down testing cycles, reducing opportunities for broader hyperparameter exploration. It also risked pipeline fragmentation (using different environments for different steps).
- **Mitigation:** Weights were converted and run using a TensorFlow variant of FinBERT, allowing reliable CPU-only inference. Batch inference and text-truncation were implemented to keep execution time manageable.

Small Dataset and Class Imbalance

- **Description:** The dataset comprises roughly one year of trading days (~252 points). Market direction labels were not evenly distributed; "up" days generally outnumbered "down" days in the sample period.
- **Impact:** This raises the risk of degenerate models—for example, a model that predicts "always up" might achieve superficially high accuracy but zero predictive utility. It also limits the complexity of models that can be used without overfitting.
- **Mitigation:** Class weights were applied in the Logistic Regression to penalize the majority class. Careful fold sizing in cross-validation was used to ensure every test fold contained a mix of up/down examples.

Risk of Misalignment Between Text Sentiment and Price Labels

- **Description:** News sentiment is recorded at a daily resolution, while financial

markets process information continuously. A positive headline published late in the evening may influence the next day's price, while the same sentiment at market close influences the current day.

- **Impact:** This timing mismatch introduces noise and reduces causal interpretability.
- **Mitigation:** Daily aggregation windows were standardized to Eastern Time (ET). A 7-day rolling price average was used as the target variable to reduce sensitivity to intraday fluctuations and capture the broader "momentum" effect rather than instant reaction.

Findings and Analysis

The Model Equation: Interpreting the Signal

After training and tuning, the final Logistic Regression equation provides a transparent look into how public sentiment influences ABNB stock price. The fitted equation is:

$$y = \sigma(0.4236 * \text{title_positive} + 0.6649 * \text{title_neutral} + -1.1229 * \text{title_negative} + -0.1449 * \text{content_positive} + 0.5652 * \text{content_neutral} + -0.4546 * \text{content_negative} + -0.1175)$$

Key analytical insights:

- **The Dominance of Headlines:** The coefficient for *title_negative* is -1.12, the strongest weight in the entire model. This indicates that negative headlines are the single most significant predictor of downward price movement. This aligns with behavioral finance theories where retail investors often react impulsively to negative headlines ("panic selling") without reading the full article.
- **Neutrality as a Positive Signal:** Surprisingly, both *title_neutral* (+0.66) and *content_neutral* (+0.57) are strong positive predictors. In the context of Airbnb, this suggests that "no news is good news"—stability and a lack of controversy allow the stock's natural momentum (or broader market beta) to drive it upward.
- **Content "Noise":** The weights for content (body text) are generally lower or counter-intuitive (e.g., *content_positive* at -0.14). This implies that for short-term prediction, the article body contains more noise than signal compared to the headline.

Performance Metrics

The final model achieved an aggregated **ROC AUC of 0.552**. While this appears low compared to other domains (where 0.8+ is common), it is consistent with financial sentiment literature. Jiang and Zeng (2023) and Mittal and Goel (n.d.) often report AUCs in the 0.53–0.58 range for single-stock daily predictions.

- **Precision/Recall:** The model showed a **Precision of ~0.52** and a **Recall of ~0.65**.

This indicates the model is reasonably good at catching "Up" moves (Recall) but generates a significant number of false positives (Precision).

- **Stability:** Cross-validation folds showed variance (AUC ranging from 0.50 to 0.60), highlighting the non-stationary nature of market data.

Visual Analysis

As seen in Appendix C (Figures C3 and C4), the ROC curve lifts slightly above the diagonal, confirming the presence of a weak but non-random signal. The confusion matrices (Text C2) reveal that without class weighting, the model initially ignored "Down" days; the final tuned model successfully captures a balanced portion of both market directions.

Discussion

Evaluation Against Objectives

The project successfully achieved its core objective: building a reproducible pipeline that translates text to a market signal. The predictive performance (AUC 0.55), while modest, validates the hypothesis that public sentiment contains some predictive information, even if it is not a "crystal ball." The clear separation of "Title" vs. "Content" influence provides a granular insight often lost in aggregate scoring systems.

Deep Dive: Interpreting Model Anomalies

A critical analysis of the logistic regression weights reveals a counter-intuitive finding: *content_positive* has a negative weight (-0.14). Conventional wisdom suggests positive news should predict price increases. However, this anomaly likely points to the "Buy the Rumor, Sell the News" phenomenon. By the time positive sentiment permeates the detailed body of a news article, the market may have already priced it in, leading to profit-taking (selling) on the following day. This contrasts with *title_negative* (-1.12), which represents an immediate "shock" value that retail traders react to instantly before reading deeper.

The Signal of Stability

The strong positive weights for *title_neutral* (+0.67) and *content_neutral* (+0.57) offer a significant theoretical insight. In the context of a volatile tech stock like Airbnb, "Neutral" sentiment does not mean "no information"; it means stability. This aligns with the "Safe Harbor" hypothesis: when news is factual and non-sensational (neutral), it reduces perceived risk,

encouraging standard capital inflows and upward drift. This finding suggests that for retail investors, "boring" news days may actually be safer buying opportunities than "hype" days, as they indicate business-as-usual execution.

Comparison with Literature

The strong negative weight on headlines (-1.12) corroborates the findings of Bharathi and Geetha (2017) regarding the asymmetry of market sentiment: bad news travels faster and impacts prices harder than good news. Furthermore, the limited lift provided by the model aligns with Kirtac and Germano (2024), who warn that simple sentiment signals are increasingly arbitrated away by high-frequency algorithms. This project adds to that body of work by suggesting that the residual signal available to retail traders lies not in chasing positive hype (which the model penalizes), but in avoiding negative headlines and capitalizing on neutral stability.

Limitations

- **Data Volume:** One year of data (~252 points) is statistically small for machine learning. This limited the ability to use complex interactions or non-linear models without overfitting.
- **News Decay:** As Boyd (2019) notes, news lifecycles are short. Aggregating sentiment over 24 hours may "wash out" the immediate impact of a breaking story that occurred early in the trading day.
- **Source Bias:** Relying on Google News and Yahoo Finance excludes the raw, unfiltered sentiment found on Reddit or Twitter, which Shobayo et al. (2024) suggest

contains higher variance and potential signal.

Conclusion and Recommendations

Summary of Findings

This capstone demonstrates that a retail-accessible pipeline using FinBERT can extract a measurable market signal from public news. The most valuable finding is the "Headline Risk" factor: negative titles are the strongest predictor of price declines, while article content is largely secondary. The final model provides a baseline AUC of 0.55, establishing a "floor" for performance that confirms feasibility but suggests room for improvement.

Recommendations for Practitioners

- **Focus on Headlines:** For retail traders building similar tools, computational resources should be prioritized on processing headlines in near real-time rather than full article bodies.
- **Expand Data Sources:** Integrate high-frequency social data (Twitter/X, Reddit) to capture the "pre-news" chatter, as suggested by Mittal and Goel (n.d.).

Future Work

Future iterations should explore Hybrid Models (Shobayo et al., 2024), combining the interpretability of Logistic Regression with the feature extraction power of LLMs (e.g., GPT-4) to summarize narrative themes, not just sentiment scores. Additionally, implementing SHAP (Shapley Additive exPlanations) would provide a day-by-day breakdown of which specific news articles contributed to a buy/sell signal, increasing trust for the end-user.

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Appendix A — WBS & Gantt

Text A1: Work Breakdown Structure

1. Initiation

1.1. Define Project Scope

- ◆ Specify final prediction target: up/down next-day ABNB close.
- ◆ Define exact dataset: Jan 1–Dec 31, 2024, online content and ABNB OHLC data from Yahoo Finance.
- ◆ Document pipeline steps: scraping → sentiment → features → model → evaluation → report.

1.2. Identify Stakeholders

- ◆ Record the instructor and me as sponsors.
- ◆ Record myself as the project manager and sole executor.
- ◆ Identify secondary audience: retail investors and academic evaluators.

1.3. Develop Project Charter

- ◆ Write the project charter summarizing objectives, tools, tasks, risks, and expected results.
- ◆ Save the charter in the project directory as a reference point.

2. Planning

2.1. Develop Project Plan

- ◆ Write all sections of the project plan in APA 7 format.
- ◆ Integrate the methodology justification already approved.
- ◆ Add references supporting the plan.

2.2. Define Requirements

2.2.1. Data sources

- Google News RSS feed: 20–30 articles/day
- Yahoo Finance ABNB articles
- Optional Reddit posts if accessible

2.2.2. Tools selected

- Requests + BeautifulSoup for scraping
- FinBERT via HuggingFace for sentiment
- scikit-learn logistic regression
- Matplotlib / Seaborn for plots
- FastAPI for internal structure (no deployment)

2.2.3. Define input and output file formats for each stage.

2.3. Create Schedule

- ◆ Assign tasks to weeks 1–8: initiation → scraping → sentiment → features → model → evaluation → final report.
- ◆ Create a week-based Gantt chart reflecting dependencies.

2.4. Risk Management Planning

2.4.1. Create risk log

- Insufficient daily text → add more sources

- Long FinBERT inference time → limit text length
- Poor model accuracy → adjust rolling window
- Delays → simplify scraping

3. Execution

3.1. System Design

- ◆ Create project folder structure with /app, /app/models, /data, /data/ABNB, /data/ABNB/raw_text, /outputs.
- ◆ Build a FastAPI skeleton to organize internal scripts into clear endpoints (not deployed).
- ◆ Define modular scripts: main.py, sentiment_analysis.py, technical_analysis.py, logistic_regression.py, and evaluate.py.

3.2. Development

3.2.1. Scraping

- Develop a script to query Google News RSS for "Airbnb" daily.
- Extract titles, timestamps, and article summaries.
- Clean text: remove boilerplate, ads, duplicates.
- Save each day's results in a JSON file.

3.2.2. Sentiment Scoring

- Load FinBERT and run sentiment inference on each text item.
- Aggregate daily sentiment: $\text{mean}(P(\text{pos}) - P(\text{neg}))$.
- Store daily sentiment values in CSV.

3.2.3. Feature Engineering

- Combine daily sentiment with ABNB stock data (downloaded via yfinance).
- Generate rolling features: sentiment_t-1..t-7, rolling mean, rolling std.
- Compute next-day direction label (1 = up, 0 = down).

3.2.4. Modeling

- Train logistic regression using a chronological split.
- Save the trained model and metrics.
- Produce prediction plots and a confusion matrix.

3.3. Testing

- ◆ Validate text scraping results with manual inspection.
- ◆ Check sentiment output distributions to ensure FinBERT is behaving correctly.
- ◆ Confirm chronological separation in training/testing sets.

3.4. Implementation

- ◆ Produce all plots for the final paper: sentiment trend, rolling features, and confusion matrix.
- ◆ Write interpretation sections explaining feature influence and prediction behavior.
- ◆ Prepare final, organized dataset and model outputs.

4. Monitoring and Controlling

4.1. Track Progress

- ◆ Weekly self-review comparing completed tasks to the Gantt.
- ◆ Adjust workload if scraping or model training is delayed.

4.2. Quality Control

- ◆ Inspect raw text for major errors.
- ◆ Review sentiment data for anomalies (zeros, NaNs).
- ◆ Confirm logistic regression coefficients are stable and interpretable.

4.3. Risk Monitoring

- ◆ Update risk log weekly.
- ◆ If scraping fails, reduce to one primary source (e.g., Yahoo Finance).
- ◆ If features are sparse, shorten the rolling window to 3 days.

5. Closing

5.1. Final Review

- ◆ Make sure all pipeline steps are complete and reproducible.
- ◆ Ensure all plots, tables, and metrics are finalized.

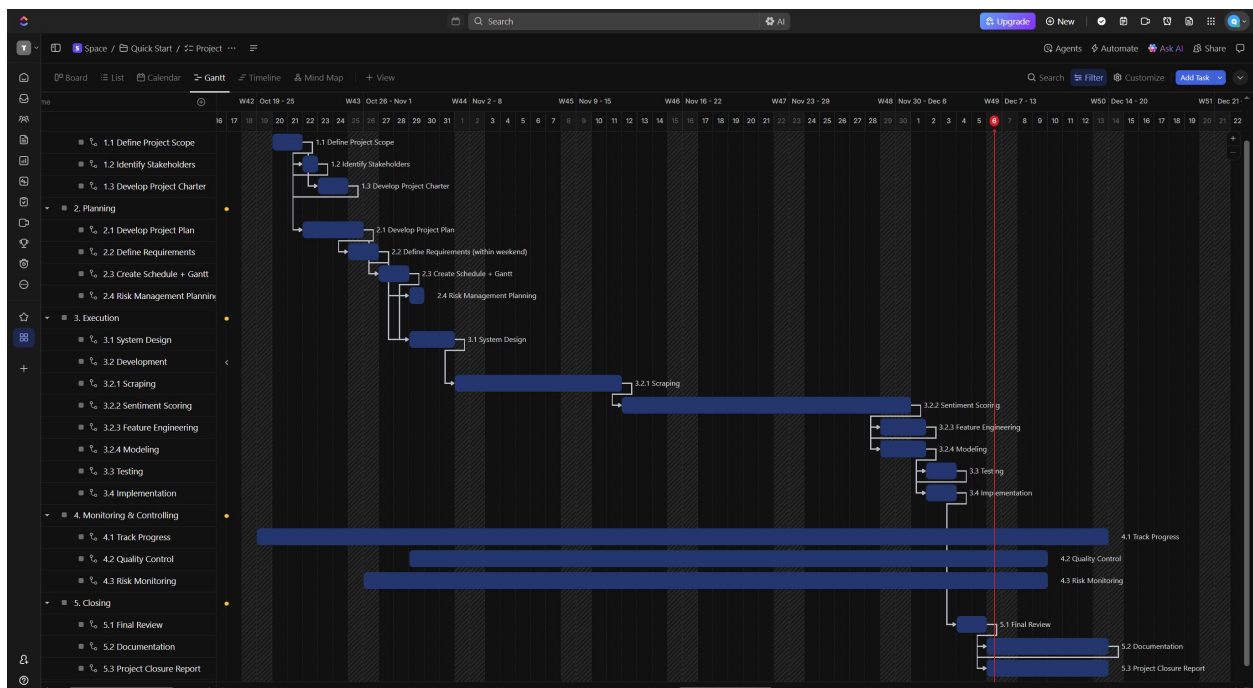
5.2. Documentation

- ◆ Write the final capstone report, including methodology, results, and limitations.
- ◆ Prepare appendices with sample scraped text, feature descriptions, and model outputs.

5.3. Project Closure Report

- ◆ Document lessons learned following guidance from Guthrie (2024) and Roy (2025).
- ◆ Summarize what worked well, what failed, and what could improve future versions.
- ◆ Store closure report in the project directory for submission.

Figure A1: Gantt Chart



Appendix B — Scraped Text & Provenance

File B1: “data/ABNB/news/2024-03-11.json” (has missing content & potential duplicates)

```
[
  {
    "date": "2024-03-11",
    "title": "Airbnb to ban indoor security cameras in rentals",
    "url":
"https://seekingalpha.com/news/4078259-airbnb-ban-indoor-security-cameras-rentals",
    "source": "seekingalpha.com",
    "content": ""
  },
  {
    "date": "2024-03-11",
    "title": "Airbnb bans the use of indoor security cameras",
    "url":
"https://edition.cnn.com/2024/03/11/tech/airbnb-bans-the-use-of-indoor-security-cameras/",
    "source": "edition.cnn.com",
    "content": "Airbnb is banning the use of indoor security cameras in listings globally, the short-term rental platform announced Monday.\n\nPreviously, Airbnb (ABNB) allowed hosts to have indoor security cameras in common areas as long as the devices were clearly disclosed on the listing page and placed in visible spots in the home. Indoor security cameras, however, were never allowed in spaces like sleeping areas and bathrooms.\n\nThe update announced Monday clarifies that indoor security cameras are now prohibited in any listing, regardless of their location, purpose or prior disclosure.\n\n“Our goal was to create new, clear rules that provide our community with greater clarity about what to expect on Airbnb,” Juniper Downs, Airbnb’s head of community policy and partnerships, said in a company blog post Monday.\n\nDowns added that the new policy was created in consultation with guests, hosts and privacy experts.\n\nThe company emphasized that the majority of listings on Airbnb currently do not report having a security camera, meaning the policy update should only impact a small subset of hosts.\n\nHosts who do currently have indoor security cameras have until April 30 to remove them. After this date, a host who violates the new policy could face consequences including listing or account removal on the platform.\n\nOutdoor security cameras, such as doorbell cameras, will continue to be permitted in Airbnb. Hosts, however, will be required to disclose the presence and general location of any outdoor cameras to guests before they book."
  },
  {
    "date": "2024-03-11",
```

```

    "title": "Airbnb bans the use of indoor security cameras",
    "url":
"https://abc13.com/airbnb-indoor-security-cameras-host-changes-doorbell-cams/14514564
/",
    "source": "abc13.com",
    "content": "NEW YORK -- Airbnb is banning the use of indoor security cameras in
listings globally, the short-term rental platform announced Monday.\n\nPreviously, Airbnb
(ABNB) allowed hosts to have indoor security cameras in common areas as long as the devices
were clearly disclosed on the listing page and placed in visible spots in the home. Indoor
security cameras, however, were never allowed in spaces like sleeping areas and
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will continue to be permitted in Airbnb. Hosts, however, will be required to disclose the
presence and general location of any outdoor cameras to guests before they book."
    },
    {
        "date": "2024-03-11",
        "title": "Airbnb bans the use of indoor security cameras",
        "url":
"https://abc7.com/airbnb-indoor-security-cameras-host-changes-doorbell-cams/14514564/
",
        "source": "abc7.com",
        "content": "NEW YORK -- Airbnb is banning the use of indoor security cameras in
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greater clarity about what to expect on Airbnb,\" Juniper Downs, Airbnb's head of community
policy and partnerships, said in a company blog post Monday.\n\nDowns added that the new

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policy was created in consultation with guests, hosts and privacy experts.\n\nThe company emphasized that the majority of listings on Airbnb currently do not report having a security camera, meaning the policy update should only impact a small subset of hosts.\n\nHosts who do currently have indoor security cameras have until April 30 to remove them. After this date, a host who violates the new policy could face consequences including listing or account removal on the platform.\n\nOutdoor security cameras, such as doorbell cameras, will continue to be permitted in Airbnb. Hosts, however, will be required to disclose the presence and general location of any outdoor cameras to guests before they book."

```
    },
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    {
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        "date": "2024-03-11",
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        "title": "Airbnb bans the use of indoor security cameras",
```

```
        "url":
```

```
"https://abc7news.com/airbnb-indoor-security-cameras-host-changes-doorbell-cams/14514564/",
```

```
        "source": "abc7news.com",
```

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        "content": "NEW YORK -- Airbnb is banning the use of indoor security cameras in listings globally, the short-term rental platform announced Monday.\n\nPreviously, Airbnb (ABNB) allowed hosts to have indoor security cameras in common areas as long as the devices were clearly disclosed on the listing page and placed in visible spots in the home. Indoor security cameras, however, were never allowed in spaces like sleeping areas and bathrooms.\n\nThe update announced Monday clarifies that indoor security cameras are now prohibited in any listing, regardless of their location, purpose or prior disclosure.\n\n\"Our goal was to create new, clear rules that provide our community with greater clarity about what to expect on Airbnb,\" Juniper Downs, Airbnb's head of community policy and partnerships, said in a company blog post Monday.\n\nDowns added that the new policy was created in consultation with guests, hosts and privacy experts.\n\nThe company emphasized that the majority of listings on Airbnb currently do not report having a security camera, meaning the policy update should only impact a small subset of hosts.\n\nHosts who do currently have indoor security cameras have until April 30 to remove them. After this date, a host who violates the new policy could face consequences including listing or account removal on the platform.\n\nOutdoor security cameras, such as doorbell cameras, will continue to be permitted in Airbnb. Hosts, however, will be required to disclose the presence and general location of any outdoor cameras to guests before they book."
```

```
    },
```

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    {
```

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        "date": "2024-03-11",
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        "title": "Airbnb bans the use of indoor security cameras",
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```
        "url":
```

```
"https://www.erienewsnow.com/story/50554889/airbnb-bans-the-use-of-indoor-security-cameras",
```

```

        "source": "erienewsnow.com",
        "content": ""
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        "date": "2024-03-11",
        "title": "Airbnb bans the use of indoor security cameras - LocalNews8 . com",
        "url":
"https://localnews8.com/news/national-world/2024/03/11/airbnb-bans-the-use-of-indoor-
security-cameras/",
        "source": "localnews8.com",
        "content": "By Catherine Thorbecke, CNN\n\nNew York (CNN) – Airbnb is banning the
use of indoor security cameras in listings globally, the short-term rental platform
announced Monday.\n\nPreviously, Airbnb (ABNB) allowed hosts to have indoor security
cameras in common areas as long as the devices were clearly disclosed on the listing page
and placed in visible spots in the home. Indoor security cameras, however, were never
allowed in spaces like sleeping areas and bathrooms.\n\nThe update announced Monday
clarifies that indoor security cameras are now prohibited in any listing, regardless of
their location, purpose or prior disclosure.\n\n“Our goal was to create new, clear rules
that provide our community with greater clarity about what to expect on Airbnb,” Juniper
Downs, Airbnb’s head of community policy and partnerships, said in a company blog post
Monday.\n\nDowns added that the new policy was created in consultation with guests, hosts
and privacy experts.\n\nThe company emphasized that the majority of listings on Airbnb
currently do not report having a security camera, meaning the policy update should only
impact a small subset of hosts.\n\nHosts who do currently have indoor security cameras
have until April 30 to remove them. After this date, a host who violates the new policy
could face consequences including listing or account removal on the platform.\n\nOutdoor
security cameras, such as doorbell cameras, will continue to be permitted in Airbnb. Hosts,
however, will be required to disclose the presence and general location of any outdoor
cameras to guests before they book.\n\nThe use of indoor security cameras in Airbnb listings
has long drawn frustration from travelers using the platform. The update on Monday also
comes approximately a week after NBC’s “Saturday Night Live” aired a skit that made jokes
about cameras in the toilet of an Airbnb listing."
    }
]

```

File B2: “data/ABNB/news/2024-07-12.json” (has no usable article for sentiment analysis)

```

[
    {
        "date": "2024-07-12",
        "title": "AIA Group Ltd Has $3 . 65 Million Holdings in Airbnb , Inc . ( NASDAQ :
ABNB ) ",
    }
]

```

```

        "url":
"https://www.dailypolitical.com/2024/07/12/aia-group-ltd-has-3-65-million-holdings-in
-airbnb-inc-nasdaqabnb.html",
        "source": "dailypolitical.com",
        "content": ""
    }
]

```

File B3: "data/ABNB/news/2024-12-13.json" (a perfect sample)

```

[
  {
    "date": "2024-12-13",
    "title": "Travel Troubleshooter : Airbnb rental is located in a high - crime ,
dodgy neighborhood",
    "url":
"https://www.orlandosentinel.com/2024/12/09/travel-troubleshooter-airbnb-rental-is-lo
cated-in-a-high-crime-dodgy-neighborhood/",
    "source": "orlandosentinel.com",
    "content": "DEAR TRAVEL TROUBLESHOOTER: I had a long-term Airbnb booked in
Washington, D.C., but my host canceled. Airbnb helped me rebook another location with a
slight discount. I was just checking the new area out a few days ago, when I realized that
it is in a really dangerous area with super-bad murder and auto theft rates, as well as
news articles mentioning my street by name.\n\nI brought all this up with Airbnb's support
staff and got the response that Airbnb doesn't really care about the locality, just the
place. There have been multiple murders on my street, and I can't park at the place. I
have to park down the street, so the locality is important as I have to walk through it
each day.\n\nI feel unsafe, and I really want to get a refund. I don't think it's fair
that Airbnb offers a place like this. I also don't think Airbnb should have sent me this
listing after my first rental fell through. Can you help me?\n\n- Robert Harris,
Harrisonburg, Virginia\n\nANSWER: Airbnb shouldn't have stuck you with a rental in a
high-crime neighborhood. According to its published policy, if your host cancels, it will
help you find "a similar place." But that was not a similar place.\n\nHere's where things
went sideways. An Airbnb representative offered to accommodate you, and you accepted the
new rental. You could have researched the neighborhood and crime statistics before you
said "yes" - not afterward.\n\nSome Airbnb apartments won't disclose the address until
after you've rented. That's fine. I think if you'd said something shortly after you decided
on the new place, Airbnb would have been able to help. Some Airbnbs have a generous
cancellation policy, which would have allowed you to cancel your rental and receive a full
refund.\n\nI also agree with you that Airbnb should not offer a home in an unsafe
neighborhood. And if people were being murdered on your street, I would say this is a red
flag.\n\nI reviewed the paper trail between you and Airbnb. It's difficult to read your

```

interactions. You repeatedly state that you feel unsafe and would like a refund. An Airbnb “ambassador” – the word they use for a customer service agent – then repeats what you said and dismisses your complaint.\n\nYou furnished Airbnb with ample evidence that the rental was in a dangerous neighborhood. I can’t believe it insisted on keeping your reservation. (Technically, it said you could get a partial refund and that it was up to your host, but still.)\n\nAn appeal to one of the Airbnb executives might have helped. I publish the names, numbers and email addresses of the Airbnb customer service managers on my consumer advocacy site, Elliott.org.\n\nI love the way this case was resolved. You finally reached Airbnb by phone and talked them into a partial refund. I also contacted Airbnb on your behalf, and it upped the offer to a full refund of the \$1,500 you had spent on the apartment."

},

{

"date": "2024-12-13",

"title": "12 Airbnbs that make the holidays merrier",

"url": "https://www.hospitalitynet.org/news/4124952.html",

"source": "hospitalitynet.org",

"content": "Key Takeaways\n\nFor all those still searching for a Santa-friendly stay this season, Airbnb is sharing 12 Santa-ready homes around the world.\n\nMany homes on Airbnb come with a kitchen to bake cookies, a mantle for stockings, room for a tree, or – most importantly – a fully functioning chimney.\n\nMore families are turning to Airbnb to get away, as stays on Airbnb can offer the ability to stay in the location they want, at a better value.\n\nWith the holiday season officially upon us, more families¹ are turning to Airbnb to get away – in fact, family travel grew 15 percent since last year², in part because stays on Airbnb can offer families the ability to stay in the location they want³, at a better value⁴.
 \n\nBut for believers who will be away for the holidays, there’s another perk to booking a stay on Airbnb over a hotel – homes are much preferred for visits from Santa Claus, making it as easy as possible for him to deliver presents to all those on his “nice” list. After all, many homes on Airbnb come with a kitchen to bake cookies, a mantle for stockings, room for a tree, or – most importantly – a fully functioning chimney.\n\nFor all those still searching for a Santa-friendly stay this season, Airbnb is today sharing 12 Santa-ready homes around the world. From a Christmas tree farm to a cabin right in the North Pole, these unique stays will make any holiday merrier.\n\nAll homes in this article are intended purely to inspire and illustrate. Airbnb does not recommend or endorse these listings or any other homes on the platform.\n\n¹. Family travel is defined by bookings or searches that involve one or more baby/child.\n\n². Based on global nights stayed by families from Jan. 1 – Jun. 30, 2024 compared to the same period in 2023.\n\n³. Based on feedback from more than 110,000 Airbnb guests globally surveyed between Dec. 1, 2023 – Nov. 2024.\n\n⁴. When comparing Airbnb’s Q2 2024 ADR for two bedroom, two bathroom listings to an average ADR for Marriott, Hyatt and Hilton based on publicly available Q2 2024 financial reporting, families can get an Airbnb with two bedrooms and two bathrooms for approximately the price of one hotel room at a major chain.\n\nAbout

Airbnb\n\nAirbnb was born in 2007 when two hosts welcomed three guests to their San Francisco home, and has since grown to over 5 million hosts who have welcomed over 2 billion guest arrivals in almost every country across the globe. Every day, hosts offer unique stays and experiences that make it possible for guests to connect with communities in a more authentic way.\n\nAbout Airbnb.org\n\nAirbnb.org is a nonprofit organization dedicated to facilitating temporary stays for people in times of crisis around the world. Airbnb.org operates independently and leverages Airbnb, Inc.'s technology, services, and other resources at no charge to carry out Airbnb.org's charitable purpose. The inspiration for Airbnb.org began in 2012 with a single host named Shell who opened up her home to people impacted by Hurricane Sandy. This sparked a movement and marked the beginning of a program that allows hosts on Airbnb to provide stays for people in times of need. Since then, the program has evolved to focus on emergency response and to help provide stays to evacuees, relief workers, refugees, asylum seekers, and frontline workers fighting the spread of COVID-19. Since then, hosts have offered to open up their homes and helped provide accommodations to 100,000 people in times of need. Airbnb.org is a separate and independent entity from Airbnb, Inc. Airbnb, Inc. does not charge service fees for Airbnb.org supported stays on its platform.\n\nAirbnb Press\n\nAirbnb Press Office\n\nAirbnb"

}

]

Appendix C — Model Metrics & Plots

Text C1: First Turn Metrics

Equation:

```
y = sigmoid(0.4311*title_positive + 0.6736*title_neutral +
-1.1210*title_negative + -0.1340*content_positive +
0.5691*content_neutral + -0.4515*content_negative + -0.0108)
```

Per-fold metrics:

```
{'fold': 0, 'precision': 0.47540983606557374, 'recall': 1.0, 'f1':
0.6444444444444445, 'roc_auc': 0.5985991379310345, 'pr_auc':
0.6025895976303443, 'log_loss': 0.7442396873112159, 'confusion_matrix':
[[0, 32], [0, 29]]}, {'fold': 1, 'precision': 0.5833333333333334, 'recall':
0.9722222222222222, 'f1': 0.7291666666666666, 'roc_auc': 0.55, 'pr_auc':
0.6231220212148918, 'log_loss': 0.6770244012794577, 'confusion_matrix':
[[0, 25], [1, 35]]}, {'fold': 2, 'precision': 0.3620689655172414, 'recall':
1.0, 'f1': 0.5316455696202531, 'roc_auc': 0.45773809523809517, 'pr_auc':
0.30026253273148623, 'log_loss': 0.7415439225249605, 'confusion_matrix':
[[3, 37], [0, 21]]}, {'fold': 3, 'precision': 0.7027027027027027, 'recall':
0.65, 'f1': 0.6753246753246753, 'roc_auc': 0.5499999999999999, 'pr_auc':
0.7233714735166665, 'log_loss': 0.6899358391213546, 'confusion_matrix':
[[10, 11], [14, 26]]}, {'fold': 4, 'precision': 0.5217391304347826,
'recall': 0.8275862068965517, 'f1': 0.64, 'roc_auc': 0.591594827586207,
'pr_auc': 0.4961309847313651, 'log_loss': 0.6840030091113423,
'confusion_matrix': [[10, 22], [5, 24]]}]
```

Aggregated metrics:

```
{'precision': 0.5152671755725191, 'recall': 0.8709677419354839, 'f1':
0.6474820143884892, 'roc_auc': 0.4844731182795699, 'pr_auc':
0.5141821904172766, 'log_loss': 0.7073493718696662, 'confusion_matrix':
[[23, 127], [20, 135]]}
```

Figure C1: First Turn ROC Curve

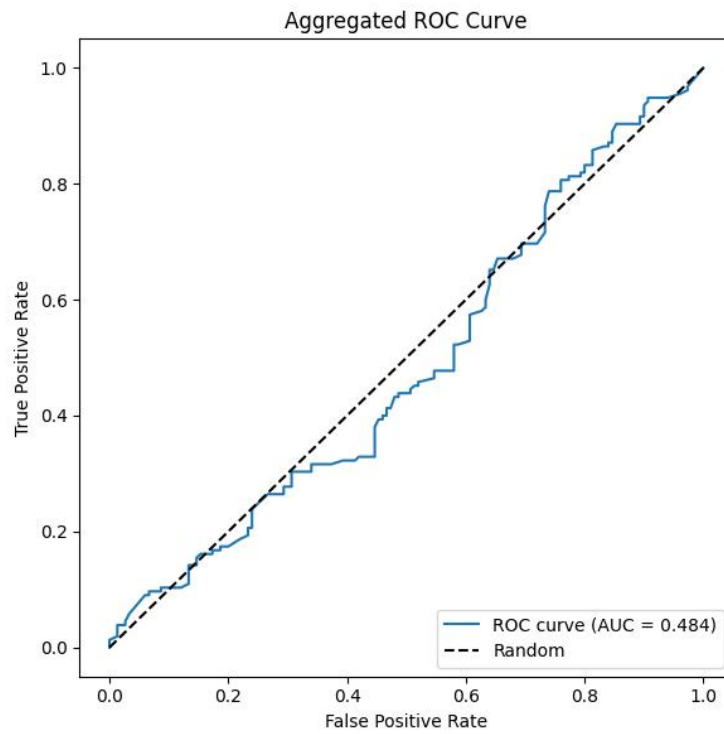
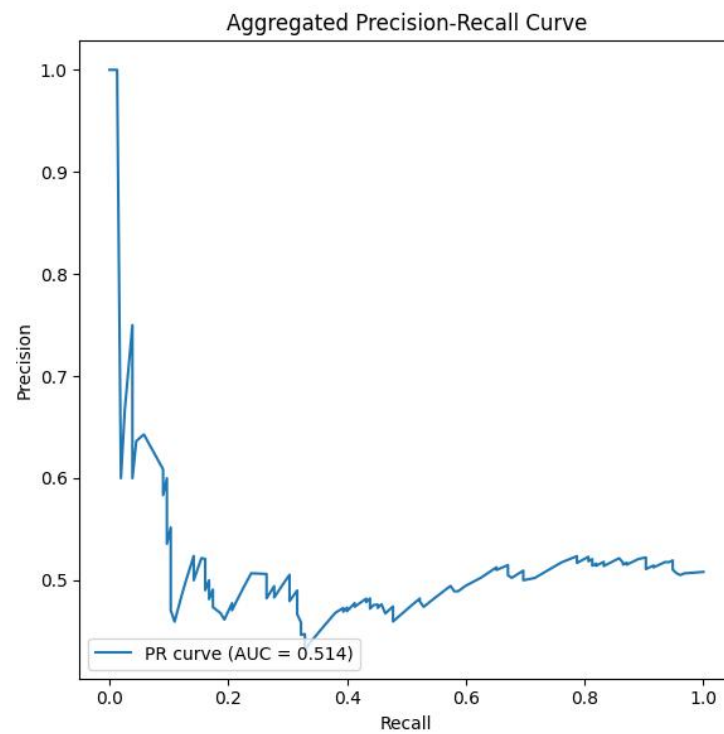


Figure C2: First Turn PR Curve



Text C2: Final Turn Metrics

Equation:

$$y = \text{sigmoid}(0.4236 \cdot \text{title_positive} + 0.6649 \cdot \text{title_neutral} + \\ -1.1229 \cdot \text{title_negative} + -0.1449 \cdot \text{content_positive} + \\ 0.5652 \cdot \text{content_neutral} + -0.4546 \cdot \text{content_negative} + -0.1175)$$

Per-fold metrics:

```
{'fold': 0, 'precision': 0.5405405405405406, 'recall':
0.8333333333333334, 'f1': 0.6557377049180327, 'roc_auc':
0.602470930232558, 'pr_auc': 0.6164688448633925, 'log_loss':
0.6874337597172943, 'confusion_matrix': [[9, 34], [8, 40]]}, {'fold': 1,
'precision': 0.39622641509433965, 'recall': 0.5384615384615384, 'f1':
0.45652173913043476, 'roc_auc': 0.5017258382642998, 'pr_auc':
0.4255694171114769, 'log_loss': 0.6872680094165203, 'confusion_matrix':
[[20, 32], [18, 21]]}, {'fold': 2, 'precision': 0.6511627906976745,
'recall': 0.56, 'f1': 0.6021505376344086, 'roc_auc': 0.5617073170731708,
'pr_auc': 0.5545350927029189, 'log_loss': 0.6907930758800157,
'confusion_matrix': [[26, 15], [22, 28]]}]
```

Aggregated metrics:

```
{'precision': 0.5235294117647059, 'recall': 0.6496350364963503, 'f1':
0.5798045602605864, 'roc_auc': 0.5515510948905109, 'pr_auc':
0.5305040477241203, 'log_loss': 0.6884982816712768, 'confusion_matrix':
[[55, 81], [48, 89]]}
```

Figure C3: Final Turn ROC Curve

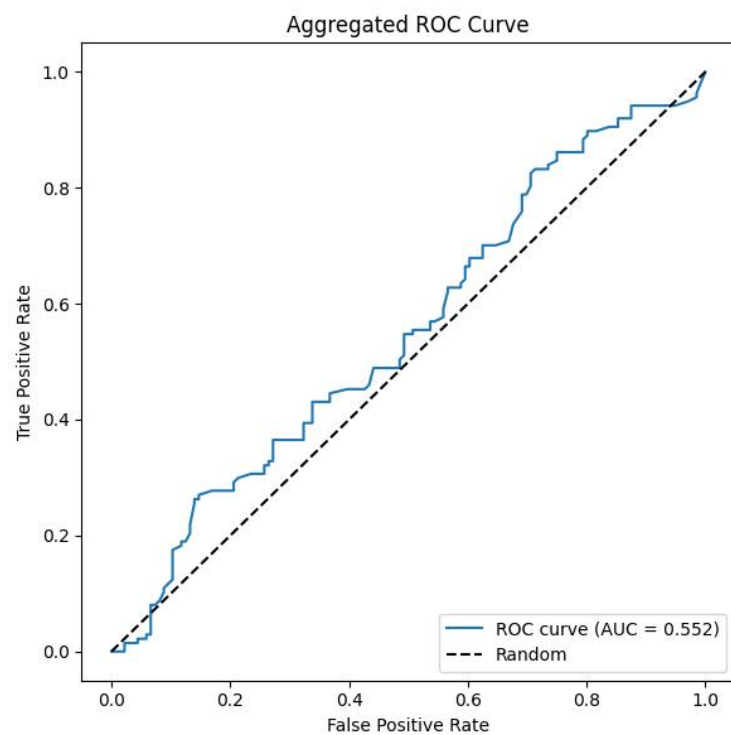


Figure C4: Final Turn PR Curve

